

Customer Churn Prediction Platform

Tech Stack: Python, Scikit-learn, Pandas, NumPy, XGBoost

- Developed an end-to-end customer churn prediction system using supervised machine learning techniques.
 - Performed extensive EDA, preprocessing, feature engineering, encoding, and missing value handling on customer behavioral datasets.
 - Trained and evaluated multiple classification models including:
 - Logistic Regression
 - Random Forest
 - Gradient Boosting
 - AdaBoost
 - Decision Tree
 - XGBoost
 - Achieved highest accuracy of **80.3% using Logistic Regression** after comparative model evaluation and hyperparameter experimentation.
 - Built reusable ML pipelines for preprocessing, training, evaluation, and prediction workflows.
 - Applied classification metrics including accuracy, confusion matrix, precision, recall, and F1-score for performance analysis.
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House Price Prediction System

Tech Stack: FastAPI, Streamlit, Scikit-learn, Python

- Built a machine learning-powered house price prediction platform with FastAPI backend and Streamlit frontend.
 - Implemented regression models including:
 - Linear Regression
 - Random Forest Regressor
 - XGBoost Regressor
 - Performed feature engineering, categorical encoding, outlier handling, and data normalization for improved prediction accuracy.
 - Designed REST APIs for real-time inference and integrated interactive prediction dashboards using Streamlit.
 - Structured the application using modular architecture with scalable API routing and reusable ML services.
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Walmart Sales Forecasting System

Tech Stack: Python, Pandas, Scikit-learn, Matplotlib

- Developed a retail sales forecasting system using time-series and regression-based machine learning techniques.

- Analyzed seasonal trends, store-level sales patterns, and promotional impacts using EDA and visualization techniques.
- Implemented predictive models including:
 - Linear Regression
 - Random Forest
 - XGBoost
- Performed feature engineering on temporal and transactional data to improve forecasting performance.
- Built comparative evaluation workflows to analyze model accuracy and business forecasting reliability.

HR Analytics & Employee Attrition Prediction

Tech Stack: Python, Pandas, Scikit-learn, Matplotlib, Seaborn

- Developed an HR analytics system to analyze employee attrition patterns and identify factors contributing to workforce turnover.
- Performed data preprocessing, feature engineering, exploratory data analysis (EDA), and correlation analysis on employee datasets.
- Implemented machine learning models including:
 - Logistic Regression
 - Decision Tree Classifier
- Built predictive workflows to identify high-risk employees likely to leave the organization.
- Analyzed employee behavior patterns based on factors such as job satisfaction, salary, work environment, experience level, and overtime trends.
- Generated actionable retention insights and visualization-driven reports to support HR decision-making.
- Evaluated model performance using classification metrics including accuracy, confusion matrix, precision, recall, and F1-score.
- Focused on interpretable machine learning outputs to improve business understanding and workforce retention strategies.

Industrial Equipment Monitoring & Predictive Maintenance Platform

Tech Stack: FastAPI, Streamlit, Python, Scikit-learn, Docker

- Engineered an AI-powered predictive maintenance platform for industrial equipment health monitoring and fault prediction.
- Built preprocessing and feature engineering pipelines for sensor-based industrial datasets.
- Implemented machine learning models for:
 - Fault Classification
 - Predictive Maintenance Forecasting
 - Equipment Failure Detection
- Experimented with multiple algorithms including Random Forest, XGBoost, Logistic Regression, and ensemble learning methods.

- Developed FastAPI backend services for prediction APIs, equipment history management, and model health monitoring.
 - Built interactive Streamlit dashboards for real-time equipment monitoring, analytics, and explainability visualization.
 - Designed Docker-ready modular architecture for scalable deployment and production readiness.
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Smart AI Assistant (This is MUCH stronger than simple API integration)

Tech Stack: Laravel, Python, FAISS, Vector Database, RAG, NLP, LLM APIs

- Developed an AI-powered intelligent assistant with a custom Retrieval-Augmented Generation (RAG) pipeline for contextual question answering.
- Built the complete RAG workflow from scratch including:
 - Data collection and preprocessing
 - Embedding generation
 - FAISS vector indexing
 - Semantic retrieval pipeline
 - Context-aware response generation
- Implemented vector similarity search using FAISS for efficient knowledge retrieval.
- Integrated bilingual conversational support and intelligent contextual response handling.
- Added file attachment support, document parsing, and contextual AI interactions.
- Designed backend architecture for scalable AI integration within Laravel applications.
- Worked on prompt orchestration, retrieval optimization, and response relevance improvements for enhanced user experience.

Career Guidance Engine API

Tech Stack: FastAPI, Python, RAG, FAISS, Vector Database, NLP, JWT, Pydantic, Voice APIs

- Developed an AI-powered career guidance platform providing personalized career recommendations, skill guidance, and contextual career insights through intelligent conversational APIs.
- Built a custom Retrieval-Augmented Generation (RAG) pipeline to deliver context-aware career guidance responses using semantic search and knowledge retrieval techniques.
- Implemented:
 - Context-aware search
 - Semantic retrieval
 - Vector similarity search
 - Intelligent recommendation workflows
- Created vector embeddings and integrated FAISS indexing for efficient retrieval of career-related information and contextual responses.
- Integrated voice assistant APIs for voice-enabled interaction and conversational career guidance experiences.
- Designed asynchronous REST APIs using FastAPI for scalable and high-performance request handling.

- Implemented JWT authentication, secure password hashing, request validation, and modular API architecture.
- Used Pydantic models for type-safe request/response validation and structured API contracts.
- Built interactive and developer-friendly API documentation using Swagger UI and ReDoc.
- Developed reusable backend services for recommendation pipelines, conversation handling, and contextual query processing.
- Structured the project with scalable, Docker-ready architecture and environment-based configuration management for deployment readiness.
- Focused on production-oriented backend engineering practices including exception handling, modular services, and maintainable API design.